

On-Premises to Cloud DB Migration

DATA MIGRATION STRATEGY SPECIFICATION

Project: Migration of Legacy CRM Database to AWS Aurora Target Completion Date: September 20, 2026

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1. Project Scope This document outlines the procedural strategy for migrating the legacy on-premises Oracle database (housing all historical Customer Relationship Management data) to a fully managed Amazon Aurora PostgreSQL-Compatible cluster. The database currently contains 4.2 Terabytes of data spanning 15 years of operational history.

2. Migration Methodology Due to the stringent requirement for near-zero downtime, a "Replicate and Cutover" strategy utilizing the AWS Database Migration Service (DMS) has been selected.

Phase 1: Schema Conversion (August 25 - August 30) The AWS Schema Conversion Tool (SCT) will be used to automatically translate the Oracle PL/SQL schemas, views, and stored procedures to PostgreSQL equivalents. Manual refactoring will be required for roughly 12% of the complex proprietary trigger functions.

Phase 2: Initial Full Load (September 5) A snapshot of the Oracle database will be taken. AWS DMS will initiate a full-load replication to the Aurora target. This process is expected to take approximately 48 hours. During this period, the on-premises database will remain fully operational.

Phase 3: Continuous Replication / CDC (September 7 - September 18) Once the full load completes, DMS will utilize Change Data Capture (CDC) to read the Oracle redo logs. Any transactions occurring on the source database will be asynchronously replicated to the Aurora cluster, keeping both databases in near real-time sync.

3. Cutover Window The final cutover is scheduled for Saturday, September 19, at 01:00 UTC. - The application layer will be placed in maintenance mode to sever connections to the Oracle source. - We will allow 5 minutes for the final CDC transactions to drain into Aurora. - Data validation scripts will run row-count and hash comparisons to guarantee data integrity. - DNS records will be updated to point the application connection strings to the new Aurora endpoints.